

Ch8 ARM Subroutines Quiz ANS

Q1: What is the primary purpose of the Link Register (R14)?

- A) To hold temporary data during arithmetic operations
- B) To store the return address of a subroutine call
- C) To hold the stack pointer value
- D) To store interrupt status flags

ANS: B

The Link Register (R14) stores the return address when a subroutine is called. When BL executes, it saves PC + 4 in LR, allowing the subroutine to return via BX LR.

Q2: When the BL (Branch and Link) instruction is executed, what two operations occur?

- A) PC is set to target address; LR is incremented by 1
- B) LR holds the target address; PC is incremented by 2
- C) LR = PC + 4 (return address); PC = target address
- D) SP is decremented; PC is set to target address

ANS: C

BL performs: (1) LR = PC + 4 (save return address), (2) PC = target_address (branch to subroutine).

Q3: Which instruction is used to return control to the caller from a subroutine?

- A) BL LR
- B) BX LR
- C) MOV PC, LR
- D) JMP LR

ANS: B

BX LR sets PC = LR, returning control to the caller. Equivalent to POP {PC}.

Q4: According to the ARM EABI, how are the first four 32-bit arguments passed to a subroutine?

- A) In registers R0-R3
- B) On the stack
- C) In registers R4-R7
- D) In registers R0-R3 for first two, on stack for the rest

ANS: A

ARM EABI specifies: Argument 1→R0, Argument 2→R1, Argument 3→R2, Argument 4→R3.

Q5: How is a 64-bit argument (such as a long long) passed to a subroutine?

- A) In a single 64-bit register
- B) Split across R0 and R2
- C) In two consecutive 32-bit registers (e.g., R0:R1 or R2:R3)
- D) Always passed on the stack

ANS: C

64-bit values use two consecutive registers: R0:R1 (first argument) or R2:R3 (second argument).

Q6: When a subroutine requires more than four arguments, where are the extra arguments passed?

- A) In memory-mapped registers
- B) On the stack by the caller

- C) In the program counter
- D) In the link register

ANS: B

When more than 4 arguments exist, extra arguments are passed on the stack by the caller, who is also responsible for cleaning up.

Q7: Where is a 32-bit return value placed by a subroutine?

- A) In register R0
- B) In register R7
- C) On the stack
- D) In the link register

ANS: A

32-bit return values are placed in R0. 64-bit values use R0:R1, and 128-bit values use R0-R3.

Q8: What stack convention does ARM Cortex-M use?

- A) Full ascending stack (stack grows toward high memory)
- B) Empty descending stack (stack grows toward low memory)
- C) Full descending stack (SP points to last item pushed; stack grows toward low memory)
- D) Empty ascending stack

ANS: C

ARM Cortex-M uses full descending stack: SP points to the last item pushed (full), and stack grows toward low memory addresses (descending).

Q9: PUSH {register_list} is equivalent to which instruction?

- A) STMIA SP!, {register_list}
- B) STMDB SP!, {register_list}
- C) LDMIA SP!, {register_list}
- D) LDMDB SP!, {register_list}

ANS: B

PUSH is equivalent to STMDB SP! (Store Multiple, Decrement Before). DB means decrement SP first, then store.

Q10: POP {register_list} is equivalent to which instruction?

- A) LDMIA SP!, {register_list}
- B) STMIA SP!, {register_list}
- C) LDMDB SP!, {register_list}
- D) STMDB SP!, {register_list}

ANS: A

POP is equivalent to LDMIA SP! (Load Multiple, Increment After). Load first, then increment SP.

Q11: When you execute PUSH {R1, R2, R3}, in what memory address order are the registers stored in memory?

- A) R1, R2, R3 (from lowest to highest address)
- B) R1, R2, R3 (lowest-numbered reg at lowest address; R1 pushed last)
- C) R3, R2, R1 (from lowest to highest address)
- D) No specific order; implementation-dependent

ANS: B

Registers are sorted automatically by hardware. R1 (lowest number) is stored at the lowest memory address but pushed last. Final memory: [R1] [R2] [R3] (lowest to highest).

Q12: On a PUSH operation in Cortex-M (full descending stack), when is SP decremented?

- A) After storing each register
- B) Only once at the end
- C) Before storing the first register
- D) SP is incremented, not decremented

ANS: C

In full descending stack, SP is decremented BEFORE storing (pre-decrement). For PUSH {r4, r5}, SP is decremented by 8, then stores occur.

Q13: Which statement correctly describes caller-saved and callee-saved registers?

- A) Caller-saved registers (R0-R3, R12) are preserved by the caller; callee-saved (R4-R11) by the callee
- B) All registers are callee-saved
- C) Caller-saved registers must be preserved by the callee
- D) Only R0-R3 are ever used

ANS: A

Caller-saved (R0-R3, R12): Caller preserves if needed. Callee-saved (R4-R11): Callee must preserve if modified.

Q14: If a subroutine modifies register R4-R11, what must the subroutine do?

- A) The caller is responsible for saving them
- B) The subroutine must save them (PUSH) and restore them (POP)
- C) They don't need to be preserved
- D) Save only odd-numbered registers

ANS: B

If a subroutine modifies R4-R11 (callee-saved), it must save them at entry (PUSH) and restore at exit (POP).

Q15: Why must a subroutine save the Link Register (LR) when calling another subroutine?

- A) The ARM architecture requires it
- B) The inner BL instruction overwrites LR with a new return address, losing the original return path
- C) To improve processor performance
- D) It's optional and rarely needed

ANS: B

When foo() calls bar(), the BL bar instruction overwrites LR with bar's return address, destroying foo's return address. Solution: foo must save LR before the nested call.

Q16: What defines a recursive function?

- A) A function that calls itself on smaller sub-problems to solve a larger problem, with a base case
- B) A function that calls multiple other functions
- C) A function that loops indefinitely
- D) A function with no parameters

ANS: A

A recursive function calls itself on smaller sub-problems with a base case to terminate recursion.

Q17: In a recursive factorial function, what is the base case?

- A) $\text{factorial}(n) = n \times \text{factorial}(n-1)$
- B) $\text{factorial}(0) = n$

- C) $\text{factorial}(n) = 1$ if $n \leq 1$
- D) $\text{factorial}(n) = n!7$

ANS: C

Factorial base case: if $(n \leq 1)$ return 1. Prevents infinite recursion.

Q18: In a descending stack (used by Cortex-M), toward which direction does the stack grow?

- A) Toward high memory addresses
- B) Toward low memory addresses
- C) Horizontally across the address bus
- D) In a circular pattern

ANS: B

Descending stack grows toward low memory addresses. SP decreases as items are pushed.

Q19: After executing PUSH {R1, R2}, what is the memory layout (assuming $R1=0x11111111$, $R2=0x22222222$)?

- A) [SP]: $0x11111111$, [SP+4]: $0x22222222$
- B) [SP]: $0x22222222$, [SP+4]: $0x11111111$
- C) [SP]: $0x22222222$, [SP-4]: $0x11111111$
- D) Both registers stored at the same address

ANS: A

R1 is at lower address of the new SP, R2 at higher address of SP+4.

Q20: According to the ARM EABI, what must be true about the Stack Pointer (SP/R13) after a subroutine returns?

- A) SP can be any value
- B) SP must have the same value as before the subroutine was called
- C) SP is incremented by 4
- D) SP is reset to 0

ANS: B

ARM EABI requires SP to have the same value after subroutine return as before the call. Maintains stack balance and prevents corruption.